

Cluster Analysis

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Table of Contents

1. Similarity and Dissimilarity

Defining Similarity (Between Objects)

Distance Measures

2. Hierarchical Clustering

Overview

Linkage Methods

States Example

3. Non-Hierarchical Clustering

K Means Clustering

States Example

Content adapted from:

Johnson, R. A., & Wichern, D. W. (2007). Applied Multivariate Statistical Analysis (6th ed).

Purpose of Clustering Methods

Clustering methods attempt to group (or cluster) objects based on some rule defining the similarity (or dissimilarity) between the objects.

Distinction between clustering and classification/discrimination:

- Clustering: the group labels are not known a priori
- Classification: the group labels are known (for a training sample)

The typical goal in clustering is to discover the “natural groupings” present in the data.

Table of Contents

1. Similarity and Dissimilarity

Defining Similarity (Between Objects)

Distance Measures

2. Hierarchical Clustering

Overview

Linkage Methods

States Example

3. Non-Hierarchical Clustering

K Means Clustering

States Example

What does it Mean for Objects to be “Similar”?

Let $\mathbf{x} = (x_1, \dots, x_p)^\top$ and $\mathbf{y} = (y_1, \dots, y_p)^\top$ denote arbitrary vectors.

Problem: We want some rule that measures the “closeness” or “similarity” between $\mathbf{x}$ and $\mathbf{y}$.

How we define closeness (or similarity) will determine how we group the objects into clusters.

- Rule 1: Pearson correlation between $\mathbf{x}$ and $\mathbf{y}$
- Rule 2: Euclidean distance between $\mathbf{x}$ and $\mathbf{y}$
- Rule 3: Number of matches, i.e., $\sum_{j=1}^p 1_{\{x_j=y_j\}}$

Card Clustering with Different Similarity Rules

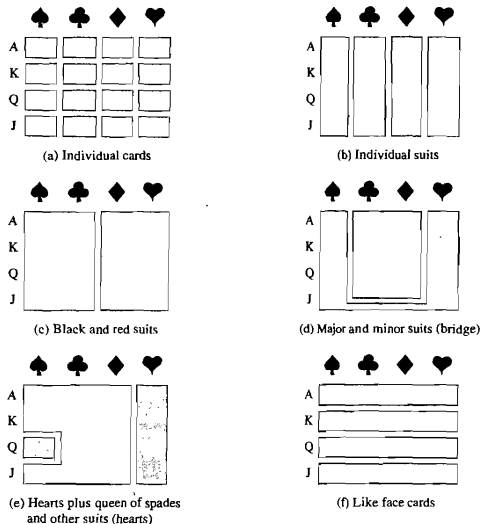


Figure 12.1 Grouping face cards.

Figure 1: Figure 12.1 from Applied Multivariate Statistical Analysis, 6th Ed (Johnson & Wichern).

Defining a Proper Distance

A **metric** (or **distance**) on a set $\mathcal{X}$ is a function $d : \mathcal{X} \times \mathcal{X} \rightarrow [0, \infty)$

Let $d(\cdot, \cdot)$ denote some **distance** measure between objects P and Q , and let R denote some intermediate object.

A proper distance measure satisfies the following properties:

1. $d(P, Q) = d(Q, P)$ [**symmetry**]
2. $d(P, Q) \geq 0$ for all P, Q [**non-negativity**]
3. $d(P, Q) = 0$ if and only if $P = Q$ [**identity of indiscernibles**]
4. $d(P, Q) \leq d(P, R) + d(R, Q)$ [**triangle inequality**]

Distances define the similarity (or dissimilarity) between objects.

Visualization of the Triangle Inequality

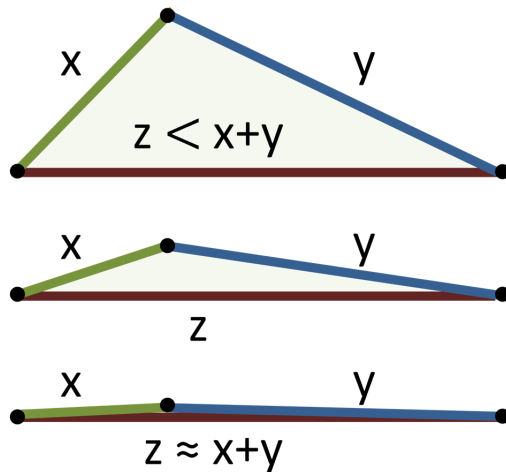


Figure 2: From https://en.wikipedia.org/wiki/Triangle_inequality

Minkowski Metric (and its Special Cases)

The **Minkowski Metric** is defined as

$$d_m(\mathbf{x}, \mathbf{y}) = \left(\sum_{j=1}^p |x_j - y_j|^m \right)^{1/m}$$

where setting $m \geq 1$ defines a true distance metric.

- Setting $m = 1$ gives the **Manhattan distance** (city block)

$$d_1(\mathbf{x}, \mathbf{y}) = \sum_{j=1}^p |x_j - y_j|$$

- Setting $m = 2$ gives the **Euclidean distance**

$$d_2(\mathbf{x}, \mathbf{y}) = \left(\sum_{j=1}^p [x_j - y_j]^2 \right)^{1/2}$$

- Setting $m = \infty$ gives the **Chebyshev distance**

$$d_\infty(\mathbf{x}, \mathbf{y}) = \max_j |x_j - y_j|$$

Table of Contents

1. Similarity and Dissimilarity

Defining Similarity (Between Objects)

Distance Measures

2. Hierarchical Clustering

Overview

Linkage Methods

States Example

3. Non-Hierarchical Clustering

K Means Clustering

States Example

Two Approaches to Hierarchical Clustering

Hierarchical clustering uses a series of successive mergers or divisions to group N objects based on some distance.

Agglomerative Hierarchical Clustering (bottom up)

1. Begin with N clusters (each object is own cluster)
2. Merge the most similar objects
3. Repeat 2 until all objects are in the same cluster

Divisive Hierarchical Clustering (top down)

1. Begin with 1 cluster (all objects together)
2. Split the most dissimilar objects
3. Repeat 2 until all objects are in their own cluster

Dissimilarity between Objects (and Clusters?)

Our input for hierarchical clustering is an $N \times N$ dissimilarity matrix

$$\mathbf{D} = \begin{pmatrix} d_{11} & d_{12} & \cdots & d_{1N} \\ d_{21} & d_{22} & \cdots & d_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ d_{N1} & d_{N2} & \cdots & d_{NN} \end{pmatrix}$$

where $d_{uv} = d(X_u, X_v)$ is the distance between objects X_u and X_v .

We know how to define dissimilarity between objects (i.e., d_{uv}), but how do we define dissimilarity between clusters of objects?

Measuring Inter-Cluster Distance (Dissimilarity)

Let $C_X = \{X_1, \dots, X_m\}$ and $C_Y = \{Y_1, \dots, Y_n\}$ denote two clusters.

- X_j is the j -th object in cluster C_X for $j = 1, \dots, m$
- Y_k is the k -th object in cluster C_Y for $k = 1, \dots, n$

To quantify the distance between two clusters, we could use:

- **Single Linkage**: minimum (or nearest neighbor) distance
$$d(C_X, C_Y) = \min_{j,k} d(X_j, Y_k)$$
- **Complete Linkage**: maximum (or furthest neighbor) distance
$$d(C_X, C_Y) = \max_{j,k} d(X_j, Y_k)$$
- **Average Linkage**: average (across all pairs) distance
$$d(C_X, C_Y) = \frac{1}{mn} \sum_{j=1}^m \sum_{k=1}^n d(X_j, Y_k)$$

Visualizing the Different Linkage Methods

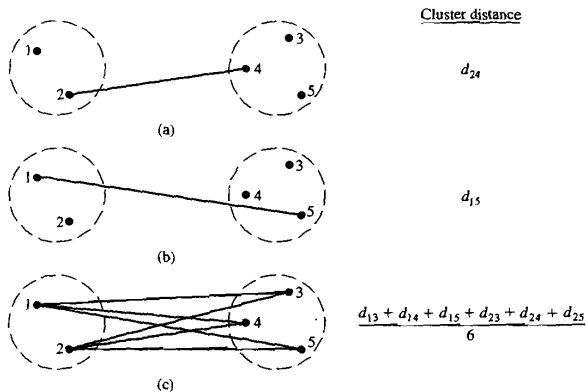


Figure 12.2 Intercluster distance (dissimilarity) for (a) single linkage, (b) complete linkage, and (c) average linkage.

Figure 3: Figure 12.2 from Applied Multivariate Statistical Analysis, 6th Ed (Johnson & Wichern)

States Example: Dissimilarity Matrix

```
# look at states data
> ?state.x77
> vars <- c("Income","Illiteracy","Life Exp","HS Grad")
> head(state.x77[,vars])
```

	Income	Illiteracy	Life Exp	HS Grad
Alabama	3624	2.1	69.05	41.3
Alaska	6315	1.5	69.31	66.7
Arizona	4530	1.8	70.55	58.1
Arkansas	3378	1.9	70.66	39.9
California	5114	1.1	71.71	62.6
Colorado	4884	0.7	72.06	63.9

```
> apply(state.x77[,vars], 2, mean)
```

	Income	Illiteracy	Life Exp	HS Grad
	4435.8000	1.1700	70.8786	53.1080

```
> apply(state.x77[,vars], 2, sd)
```

	Income	Illiteracy	Life Exp	HS Grad
	614.4699392	0.6095331	1.3423936	8.0769978

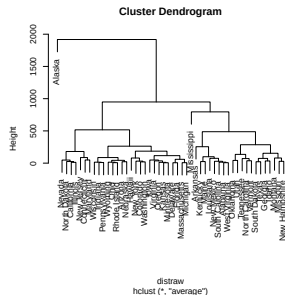
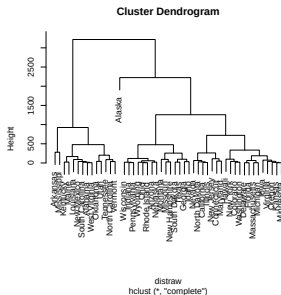
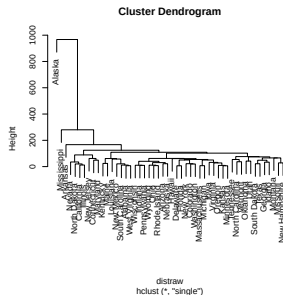
```
# create distance (raw and standarized)
> distraw <- dist(state.x77[,vars])
> diststd <- dist(scale(state.x77[,vars]))
```

States Example: HCA via Three Linkage Methods

```
# hierarchical clustering (raw data)
> hcrawSL <- hclust(distraw, method="single")
> hcrawCL <- hclust(distraw, method="complete")
> hcrawAL <- hclust(distraw, method="average")

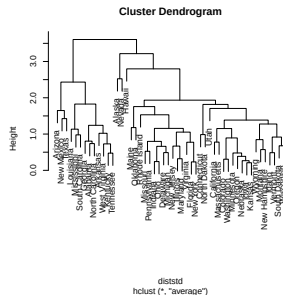
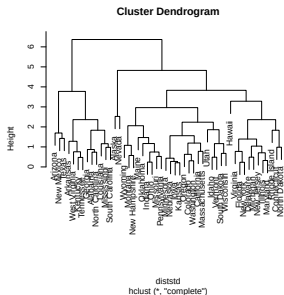
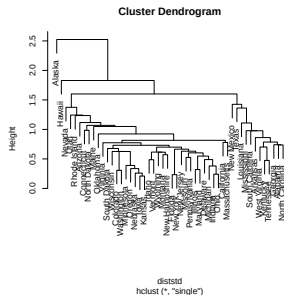
# hierarchical clustering (standardized data)
> hcstdSL <- hclust(diststd, method="single")
> hcstdCL <- hclust(diststd, method="complete")
> hcstdAL <- hclust(diststd, method="average")
```


States Example: Results for Raw Data



```
plot(hcrawSL)
plot(hcrawCL)
plot(hcrawAL)
```

States Example: Results for Standardized Data



```
plot(hcstdSL)
plot(hcstdCL)
plot(hcstdAL)
```

States Example: Standardized Data w/ Complete Link

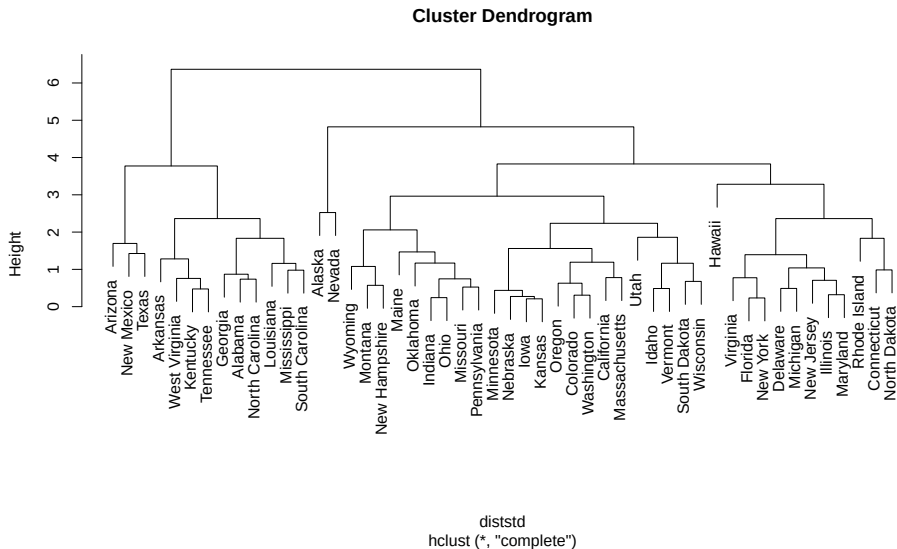


Table of Contents

1. Similarity and Dissimilarity

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Linkage Methods

States Example

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Non-Hierarchical Clustering: Definition

Non-hierarchical clustering partitions a set of N objects into K distinct groups based on some distance (or dissimilarity).

The number of clusters K can be known a priori or can be estimated as a part of the procedure.

Regardless, we need to start with some initial partition or “seed points” which define cluster centers.

- Try many different randomly generated seed points

K Means: Clustering via Distance to Centroids

K means clustering refers to the algorithm:

1. Partition the N objects into K distinct clusters $C_1, \dots, C_K$
2. For each $i = 1, \dots, N$:
 - 2a Assign object X_i to cluster C_k that has closest centroid (mean)
 - 2b Update cluster centroids if X_i is reassigned to new cluster
3. Repeat 2 until all objects remain in the same cluster

Note: we could replace step 1 with “Define K seed points giving the centroids of clusters $C_1, \dots, C_K$ ”.

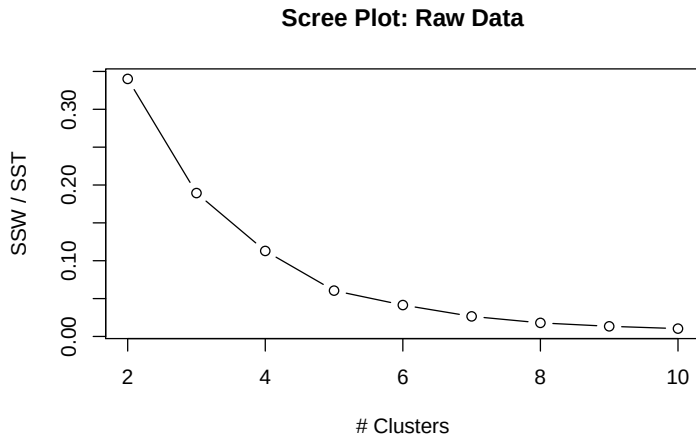
It is good to use MANY random starts of the above algorithm.

States Example: K Means on Raw Data

```
# look at states data
> ?state.x77
> vars <- c("Income","Illiteracy","Life Exp","HS Grad")
> apply(state.x77[,vars], 2, mean)
      Income Illiteracy   Life Exp    HS Grad
4435.8000      1.1700    70.8786    53.1080

# fit k means for k = 2, ..., 10 (raw data)
> kmlist <- vector("list", 9)
> for(k in 2:10){
+   set.seed(1)
+   kmlist[[k-1]] <- kmeans(state.x77[,vars], k, nstart=5000)
+ }
```

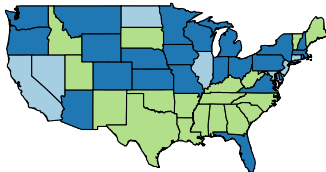
States Example: Scree Plot for Raw Data



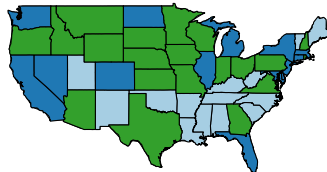
```
tot.withinss <- sapply(kmlist, function(x) x$tot.withinss)
plot(2:10, tot.withinss / kmlist[[1]]$totss, type="b", xlab="# Clusters",
     ylab="SSW / SST", main="Scree Plot: Raw Data")
```


States Example: Cluster Plot for Raw Data

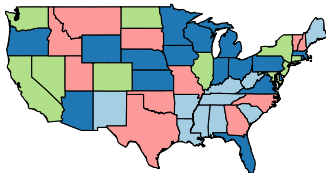
K=3 Clusters: Raw Data



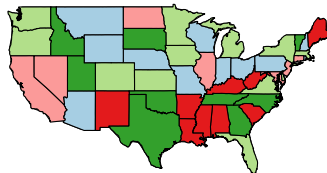
K=4 Clusters: Raw Data



K=5 Clusters: Raw Data



K=6 Clusters: Raw Data

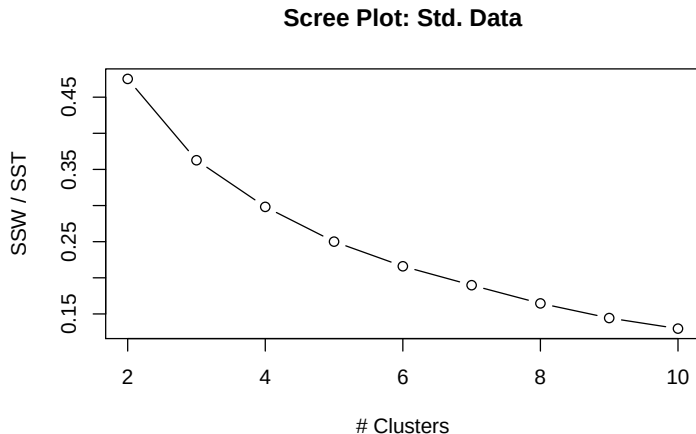


States Example: K Means on Standardized Data

```
# look at states data
> ?state.x77
> vars <- c("Income","Illiteracy","Life Exp","HS Grad")
> apply(state.x77[,vars], 2, mean)
      Income Illiteracy   Life Exp    HS Grad
4435.8000      1.1700    70.8786    53.1080

# fit k means for k = 2, ..., 10 (standardized data)
> Xs <- scale(state.x77[,vars])
> kmlist.std <- vector("list", 9)
> for(k in 2:10){
+   set.seed(1)
+   kmlist.std[[k-1]] <- kmeans(Xs, k, nstart=5000)
+ }
```

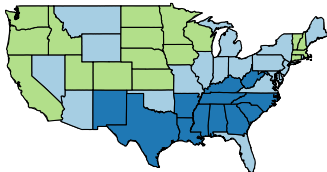
States Example: Scree Plot for Standardized Data



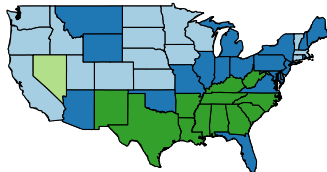
```
tot.withinss.std <- sapply(kmlist.std, function(x) x$tot.withinss)
plot(2:10, tot.withinss.std / kmlist.std[[1]]$totss, type="b",
     xlab="# Clusters", ylab="SSW / SST", main="Scree Plot: Std. Data")
```

States Example: Cluster Plot for Standardized Data

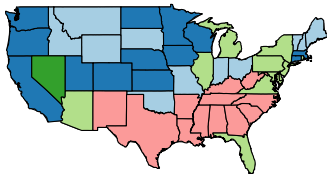
K=3 Clusters: Std. Data



K=4 Clusters: Std. Data



K=5 Clusters: Std. Data



K=6 Clusters: Std. Data

